

ELI5

Explain It Like I'm 5

Daniel Sarfaraz, Jack Owen, Kash Mulchand, Marina Prikis, Paul Garces

Background & Definition

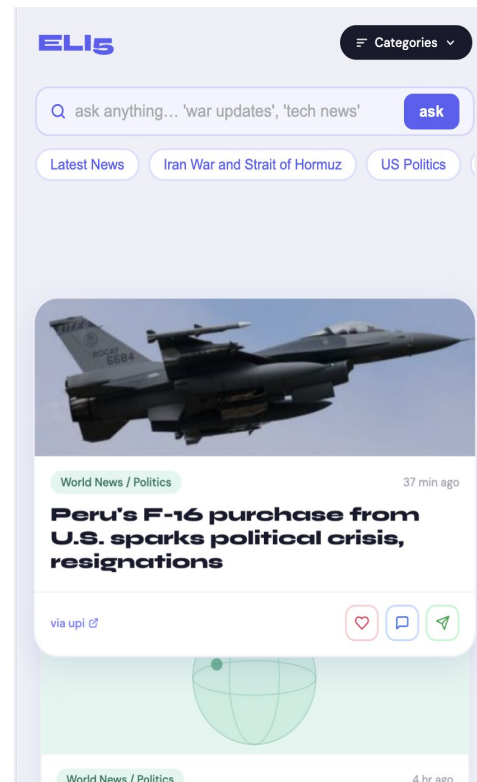
The Problem: News is fast, dense, and full of jargon. Most readers scroll past articles they don't fully understand.

The Concept: Real-time news from 3 sources, simplified by AI. Swipe, read the ELI5 version, toggle to the original, or ask follow-up questions.

Why Big Data: 20,000+ articles across 3 sources, processed with Apache Spark, and served through cloud infrastructure.

Why ELI5: 3 sources aggregated automatically, summaries pre-generated, and the ability to ask questions across thousands of articles at once.

Our competitors: New York Times, Associated Press, Morning Brew, TikTok, Instagram, and the like



Data Sources & Volume



The Guardian API

WHAT:

Full-text UK/international news; world, politics, tech, science, business

WHY:

Provides complete access to article body text, essential for ELI5 summarization

VOLUME:

Up to 200 articles per request across 7 topic categories



GNews API

WHAT:

Real-time news aggregated from 80,000+ sources across 70+ countries

WHY:

Broadest global coverage with strong topic filtering and full article content

VOLUME:

Up to 25 articles per topic, 6 topics per run, 1,000 requests/day



NewsData.io API

WHAT:

Real-time news from 50,000+ sources in 80+ languages

WHY:

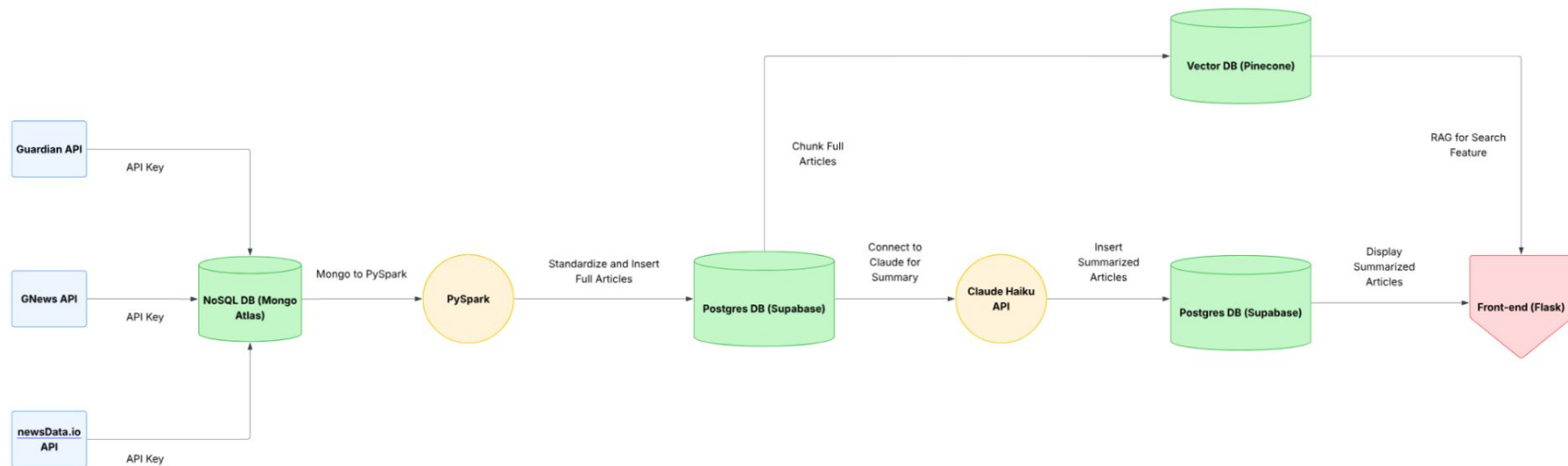
Deep category filtering, pulls from wire services like AP and Reuters, with AI summary support

VOLUME:

Up to 50 articles per request, 20,000 API credits/month across 5 categories

At Scale (Business Case): A production version would also incorporate user interaction data, reading preferences, demographic and geographic profiles, engagement metrics, and subscription/payment data to enable personalization and monetization.

Pipeline Architecture & Data Lake Diagram



Legend



Pros & Cons of Architecture

API

News APIs

PROS

Free/low cost access to real-time global news, full article body text, diverse geographic coverage

CONS

Rate limits restrict volume, inconsistent field names across sources, dependent on third-party uptime

Mo

MongoDB Atlas (NoSQL)

PROS

Schema-less handles inconsistent JSON from different APIs, flexible for unstructured data, cloud-hosted and shared across team

CONS

No joins, not ideal for structured queries, can't easily filter by multiple conditions

Py

PySpark/Databricks (Big Data)

PROS

Distributed processing scales to millions of articles, industry standard, integrates directly with Atlas

CONS

Overkill for smaller datasets, serverless cold start adds latency

PSQL

PostgreSQL/Supabase (SQL)

PROS

Structured schema enables fast filtering by category, date, and source, cloud-hosted with zero ops overhead

CONS

Requires predefined schema, doesn't handle unstructured raw data well

Pi

Pinecone (Vector DB)

PROS

Purpose-built for semantic search, fast similarity search at scale, managed cloud service

CONS

Limited to vector search only, cost increases significantly at scale

ci

Claude Haiku (AI)

PROS

Fast, cheap, high quality ELI5 summaries, simple API integration, reduces hallucinations via RAG

CONS

Cost per token adds up at scale, dependent on Anthropic's external API, no offline option

Scalability: Computation & Cost Implications

Projected infrastructure costs at 10M+ articles / 1M+ users

Tool	At Scale (10M+ articles / 1M+ users)	Cost
MongoDB Atlas	Dedicated — 10GB+, 2GB RAM	~\$57/mo
Supabase	Pro — 8GB disk, 250GB egress	~\$25/mo
Databricks	Data Engineering — \$0.15/DBU, scales with compute	variable
News APIs	Enterprise tiers across all 3 sources	~\$1,500+/mo
Kafka	Enterprise tier — high throughput, multi-zone replication	~\$895+/mo
AWS S3	Standard — 50-70GB raw storage, multi-zone, \$0.023/GB	~\$1-2/mo per 50GB
Claude Haiku	Per 10M articles summarized	~\$12,500
Render	Professional — autoscaling + global CDN	~\$19/mo
Pinecone	Standard — dedicated nodes	~\$50/mo
FastAPI	Front - end and managed DB services for scale	~\$50/mo

Data Governance



App Security

HTTPS enforced via Render deployment

API keys stored in environment variables, never in code

OAuth 2.0 user authentication via Google/GitHub, tokens managed server-side



Tool Security

MongoDB Atlas — encryption at rest & in transit, role-based access

Supabase — SSL connections, row-level security

Render — secrets managed via environment variables

AWS - Data management and recovery at scale



Data Quality

Spark pipeline enforces schema validation, null filtering, HTML stripping, and minimum word count checks on every article ingested.



Data Licensing

Guardian — Open Platform license, non-commercial use

GNews & NewsData.io — commercially licensed APIs

Attribution required for all sources per ToS

Free-tier limits support demo; paid plans needed at scale



Privacy Laws

GDPR

Applies if serving EU users, consent & data rights required

CCPA

Applies if collecting California user data, opt-out rights

HIPAA

Not applicable, no health data involved

Demo Link:

<https://eli5-news.onrender.com/>

Future Direction

Four feature expansions to increase engagement, reach, and revenue

01



Personalization

Feed adapts to reading habits and interests over time. Users onboard with topics they care about, and the more they engage, the smarter it gets.

02



Behavior-Driven Algorithm

Content ranked by engagement signals: likes, saves, and skips. Mirrors how TikTok and Reels surface relevant content without manual curation.

03



Audio & Video Formats

ELI5 summaries as 60-second audio clips or short-form video: accessible while commuting, working out, or anywhere screens aren't practical.

04



Multi-Language ELI5

Same pipeline, just a prompt change. The Guardian and GNews already pull international content, making this a low-effort, high-impact expansion.